Analysing Prediction Confidence in Support Vector Machines Using Margin Distance

MODULE: Machine Learning and Neural Networks
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ABSTRACT

Support Vector Machines (SVMs) are widely used for classification tasks due to their ability to construct optimal decision boundaries by maximising the margin between classes. However, unlike probabilistic models, SVMs do not directly provide prediction confidence.

This study investigates how prediction confidence in SVMs can be interpreted using the distance of data points from the decision boundary, referred to as margin distance. Using a synthetic dataset, the model's decision function is analysed to understand how confidence varies across samples. Visualisations and quantitative analysis are used to examine confidence distribution, misclassification patterns, and the effect of the regularisation parameter C.

The results show that samples closer to the decision boundary exhibit lower confidence and higher misclassification rates, while samples further away are classified with higher confidence. The findings demonstrate that margin distance provides a meaningful and interpretable measure of prediction confidence in SVMs.

INTRODUCTION

Machine learning models are commonly used for classification tasks in various domains such as healthcare, finance, and computer vision. In many real-world applications, it is not sufficient to produce predictions alone; understanding the confidence of those predictions is equally important.

Support Vector Machines (SVMs) are powerful classification algorithms introduced by Cortes and Vapnik (1995). They operate by constructing a decision boundary that maximises the margin between classes, leading to strong generalisation performance. However, unlike models such as logistic regression, SVMs do not naturally output probabilities, making it difficult to assess prediction confidence.

Despite this limitation, SVMs provide a decision function that reflects the distance of each data point from the decision boundary. This distance can be interpreted as a measure of confidence: points far from the boundary are classified with higher certainty, while points near the boundary are more uncertain.

This study aims to analyse prediction confidence in SVMs using margin distance. By visualising decision boundaries and examining confidence distributions, the study provides insight into how SVMs make decisions and where they are most likely to make errors.

BACKGROUND OF SVM

Support Vector Machines are supervised learning models used for classification and regression. The key objective of an SVM is to find a hyperplane that separates data points belonging to different classes while maximising the margin between them.

The margin is defined as the distance between the decision boundary and the closest data points from each class. These closest points are known as **support vectors**, and they determine the position of the decision boundary.

In non-linearly separable datasets, SVMs use kernel functions such as the Radial Basis Function (RBF) to map data into higher-dimensional space, allowing complex decision boundaries.

Margin Distance and Confidence

The SVM decision function computes a value for each data point that indicates its position relative to the decision boundary:

- Positive values indicate one class
- Negative values indicate the other class
- Values close to zero indicate proximity to the boundary

The absolute value of this function represents the distance from the decision boundary, which can be interpreted as prediction confidence:

- Large distance → high confidence
- Small distance → low confidence

DATASET AND EXPERIMENTAL SETUP

The dataset used in this study was generated using the make_moons function from the Scikit-learn library. This synthetic dataset consists of two interleaving crescent-shaped clusters, making it suitable for evaluating non-linear classification models.

Dataset Distribution

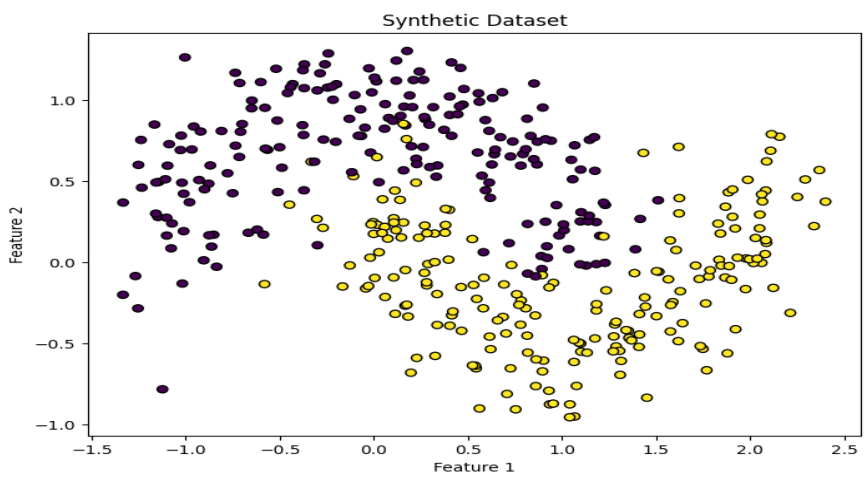

Figure 1: Synthetic dataset showing two non-linear classes.

Preprocessing

- Data split into training and testing sets (70/30)
- Feature scaling applied using StandardScaler
- Model trained using RBF kernel SVM

Model Configuration

- Algorithm: Support Vector Machine (SVM)

- Kernel: Radial Basis Function (RBF)
- Regularisation parameter $C = 1.0$

RESULTS AND VISUALISATION

1. Decision Boundary and Margins

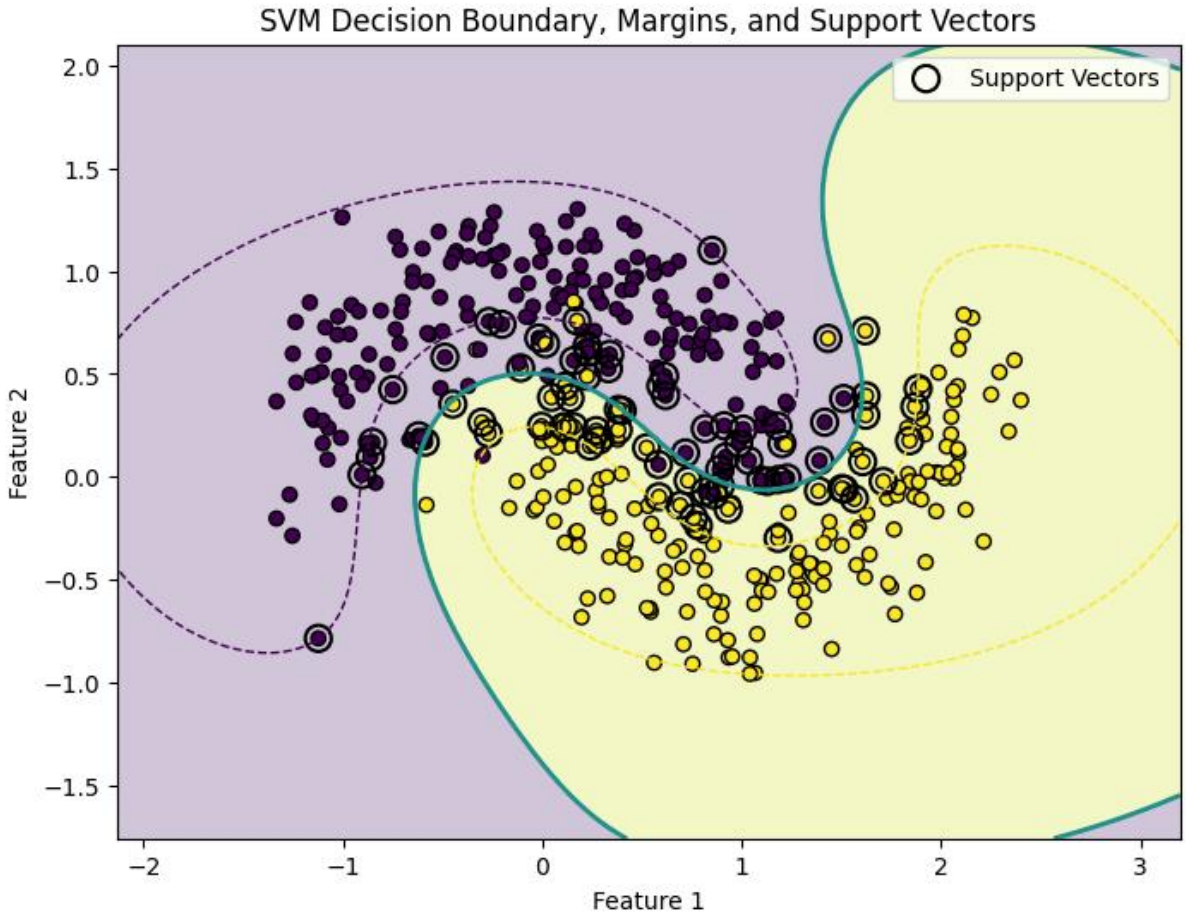

Figure 2: Decision boundary, margins, and support vectors.

The SVM constructs a non-linear boundary that separates the classes while maximising the margin. Support vectors lie near the boundary and define the separation.

2. Confidence Distribution

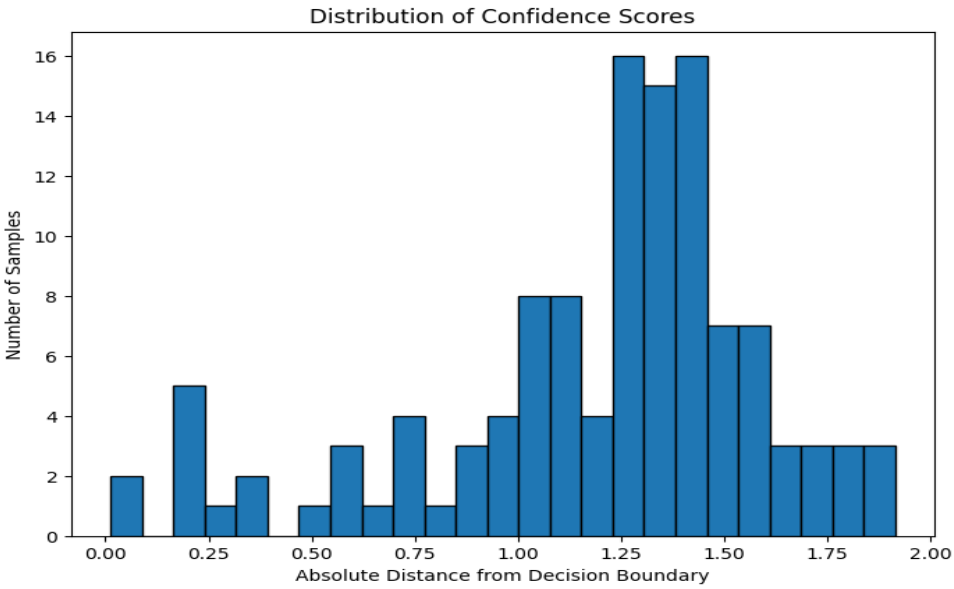

Figure 3: Distribution of confidence scores.

The histogram shows that most samples have moderate confidence, while fewer samples lie very close to the boundary.

3. Confidence-Based Visualisation

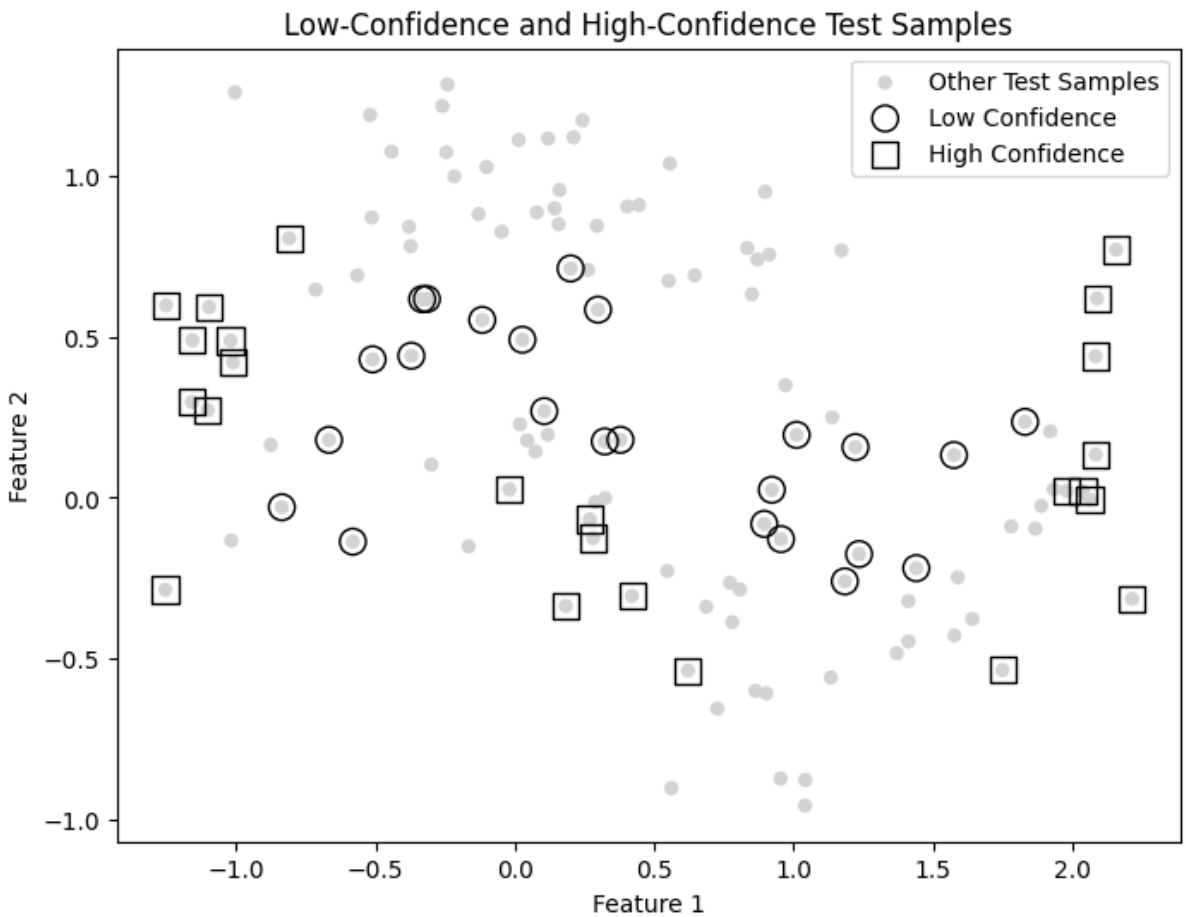

Figure 4: Confidence map across feature space.

Regions far from the boundary show high confidence, while regions near the boundary represent uncertainty.

4. Misclassification Analysis

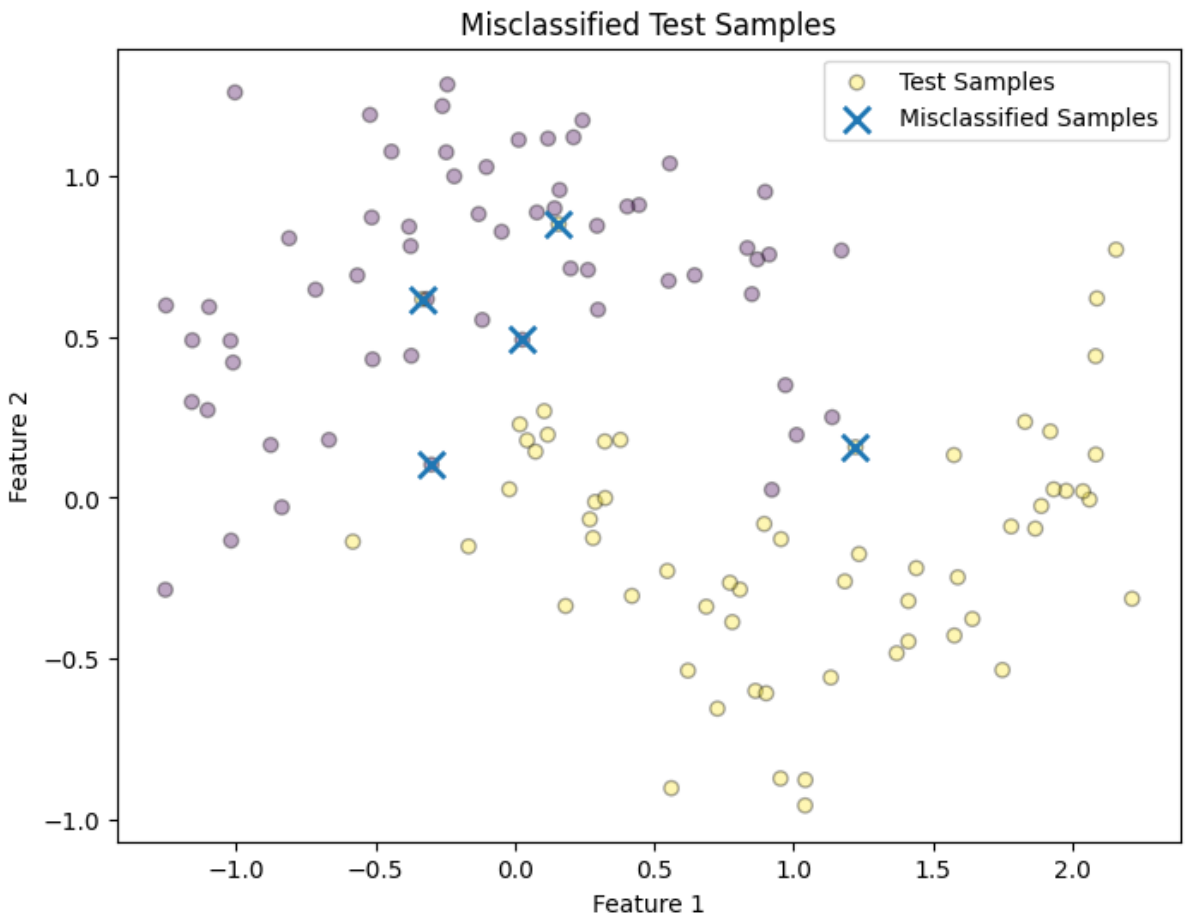

Figure 5: Misclassified samples.

Misclassified points are concentrated near the decision boundary, indicating that uncertainty is highest in this region.**5. Accuracy Across Confidence Levels**

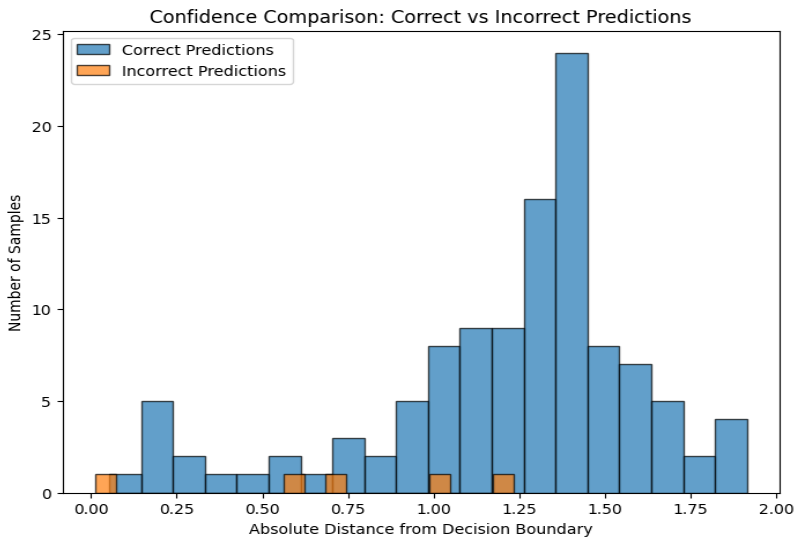

Figure 6: Accuracy across confidence bands.

Accuracy increases significantly with confidence, confirming that margin distance is a reliable indicator of prediction quality.

6. Effect of Regularisation Parameter

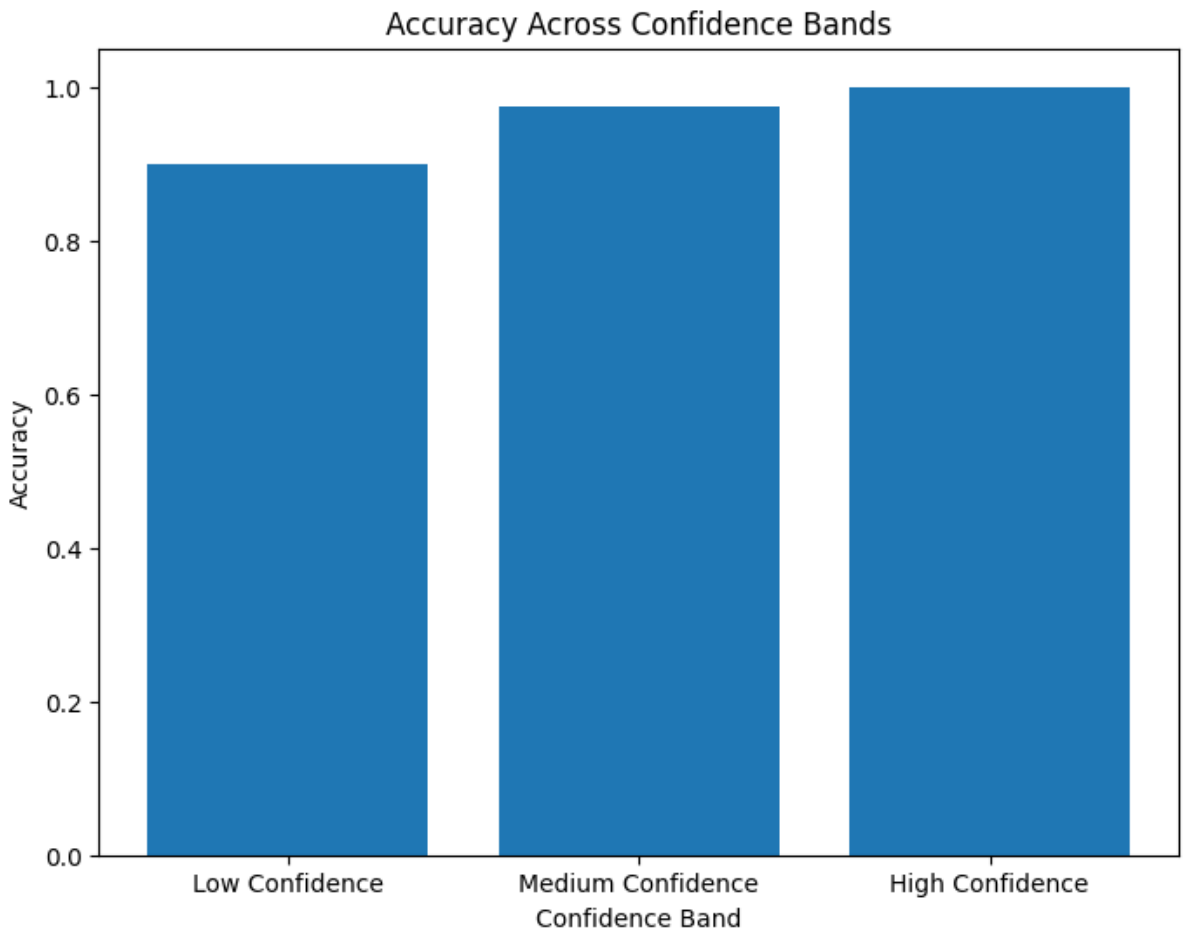

Figure 7: Effect of varying C.

Higher values of C produce more complex boundaries, while lower values produce smoother and more robust models.

DISCUSSION

The results demonstrate that margin distance provides a meaningful way to interpret SVM prediction confidence.

Samples located far from the decision boundary are classified with high confidence and are more likely to be correct. In contrast, samples near the boundary exhibit lower confidence and are more prone to misclassification.

The analysis of misclassified samples confirms this observation, as most errors occur in regions close to the boundary. This suggests that uncertainty is concentrated near the decision margin.

The study also shows that the regularisation parameter C influences both accuracy and confidence distribution. Larger values of C lead to more complex decision boundaries, which may increase overfitting and reduce robustness.

LIMITATIONS

This study has several limitations. The dataset used is synthetic and relatively simple, which may not fully represent real-world complexity. Only one kernel type was used, and alternative kernels may produce different confidence patterns. Additionally, the study does not explore probabilistic calibration methods for SVM.

FUTURE WORK

Future work could explore confidence calibration techniques such as Platt scaling or isotonic regression. More complex datasets and real-world applications could also be investigated. Additionally, comparisons with probabilistic classifiers could provide further insight into confidence estimation.

ETHICAL AND REAL-WORLD IMPLICATIONS

Understanding prediction confidence is important for real-world applications where incorrect decisions can have serious consequences. In domains such as healthcare or autonomous systems, identifying uncertain predictions can improve reliability and safety.

ACCESSIBILITY CONSIDERATIONS

All visualisations were designed with clear labels, readable fonts, and high contrast to ensure accessibility. Figures are described within the text to aid understanding.

CONCLUSION

This study analysed prediction confidence in Support Vector Machines using margin distance. The results show that the distance from the decision boundary provides a meaningful and interpretable measure of confidence.

Samples closer to the boundary are more uncertain and more likely to be misclassified, while samples further away are classified with higher confidence and accuracy. The findings highlight the usefulness of margin-based confidence for interpreting SVM models and improving their practical applicability.

REFERENCES

- Cortes, C. and Vapnik, V. (1995) *Support-vector networks*.
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